Detecting Medical Misinformation in Social Media and Headlines

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Link to Colab Notebook

Introduction

Medical misinformation threatens public health by deterring people from seeking proper medical care. The goal of this project is to build an automated classification model that accurately labels short-form medical claims as factual or fabricated. To achieve this goal, traditional machine learning is not ideal because it ignores linguistic context and word order. Misinformation is often deceptive, and a model that operates on keyword frequency will not be able to classify such text accurately. Deep learning is required because it processes text sequentially, allowing it to learn the context and grammatical patterns of deception instead of only following words (1).

Illustration/Figure

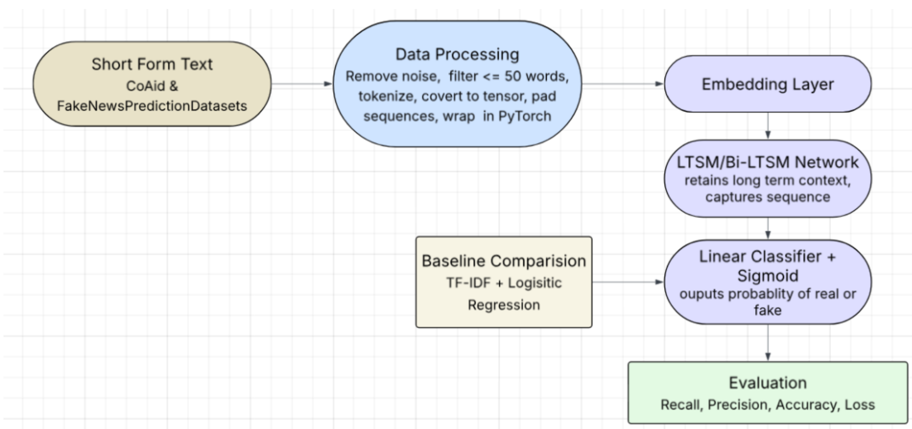

Figure 1: Illustration of the overall pipeline for the project.

Background & Related Work

Research on digital misinformation analyzes how it spreads. In their study of COVID-19 media trends, Zhao et al. (2021) define medical misinformation as false health claims that are not supported by scientific evidence (2). Zubiaga et al. (2018) similarly note in their survey that most rumours are unverified at the time of posting (3). Furthermore, Liu et al. (2019) found that misleading claims rely on personal experience and sensationalized clickbait to maximize propagation (4). Learning these patterns is crucial for training the neural network to detect misleading text accurately.

In terms of tested models, deep learning models have improved in classification accuracy in natural language processing tasks. For instance, Nasir et al. (2021) demonstrated that deep learning methods using Long Short-Term Memory (LSTM) networks can capture temporal dependencies (5). This method achieved higher detection accuracy across multiple fake news datasets compared to non-hybrid baselines. Recent advancements have focused on optimizing RNN architectures to capture contextual nuances better. Syed et al. (2023) introduced the use of Bidirectional LSTMs (BiLSTM) and an attention mechanism (1). By building on established methods, the proposed model aims to accurately classify short-form claims by identifying the underlying sequential structures of misinformation.

Data Processing

Medical claims from the CoAID and ACOVMD datasets were collected and combined into a unified dataset (6)(7). It is essential to check that all collected samples include data columns for the label ('Fake'/'Real') and the text claim for consistent formatting. The raw text was then cleaned by removing special characters and standardized casing, tokenized to a dynamic vocabulary, and padded to a 50-word sequence for uniform tensor dimensions. There are 3623 claims in the final dataset, with 1808 being real and 1815 false. The claims were then split into a 70% training set, 15% validation set, and 15% testing set. For the final test, a separate dataset with more recent samples was used (8).

Original Text	How Hillary Clinton Could Win
Cleaned & Tokenized	['hillary', 'clinton', 'could', 'win']
Label	0

Figure 2: Sample of unclean and clean data, showcasing data format (text, label).

Architecture

The proposed architecture is a custom MisinfoLSTM that allows the model to learn word order and context. The embedding layer will first convert the 50-word tokenized input into semantic vectors. The core model is a bidirectional LSTM that processes input sequences both forward and backward to capture contextual information (9). The final hidden states from both directions are concatenated into a single feature vector, which will be used as the input to the fully connected linear classification layer.

Figure 3: Pipeline of raw text tokenization, padding, and embedding vectorization.

During training, the model utilizes Binary Cross-Entropy with Logits Loss to show the model's performance. A sigmoid activation function is applied to the output logits to map the model's scores to a probability between 0 ("True") and 1 ("Fake"). The model originally used the Adam optimizer with a 0.001 learning rate, 32 batch size, and 30 epochs. However, the results show spiking validation loss due to overfitting. The current model implements several changes to address those issues. A 0.2 dropout and a dropout layer were added to prevent the neural network from focusing on specific words or phrases that appear frequently in the "Fake" category. This will force the model to look at broader sentence context and only learn the most meaningful patterns instead of memorizing key words. A weight decay was also added to prevent the model from connecting larger weights to certain vocabulary words and ensure the model does not create rigid rules that will fail on new data. These changes resulted in a validation accuracy of 95.88% and an initial test accuracy of 97.95%.

Baseline Model

A simple baseline model was created to compare to the primary model. The raw text was processed using a TF-IDF Vectorizer and passed into a Logistic Regression classifier (10). For each word in the input text, the model will assign a decimal at the corresponding index. The model will count how often specific keywords appear in a post, ignoring word order and context, making it a good baseline for comparison. The model achieved an initial accuracy of 79.37% and flagged several words as misinformation or real claims. Some of the words flagged as misinformation are “cure”, “water”, “baby”, and “depression”. The baseline model fails to detect misinformation that relies on sentence structure rather than specific trigger words, which needs to be worked on for the primary model.

Quantitative Results

The model demonstrated exceptional performance on the unseen test set, achieving an overall accuracy of 97%. It maintained balanced performance across both classes, with F1-scores of 0.98 for "True" and 0.97 for "Fake". This indicates that the model has no significant bias toward either class. An error analysis further confirms that the model achieved a False Positive Rate of 0.037 with only 5 misclassifications out of 135 actual True claims and a False Negative Rate of 0.009 with only 1 misclassification out of 109 actual Fake claims. These low error rates, combined with the high precision and recall values, demonstrate that the MisinfoLSTM architecture can effectively generalize to unseen data. The model successfully distinguishes valid medical information and misinformation while minimizing classification errors.

	Precision	Recall	F1-Score
“True”	0.97	0.99	0.98
“Fake”	0.99	0.95	0.97
Accuracy			0.97
Macro Average	0.98	0.97	0.97
Weighted Average	0.98	0.97	0.97

Figure 4: Classification Report for the Final Test Set.

Qualitative Results

The model excels at identifying "Easy" misinformation, which typically relies on hyperbolic, sensationalist language and clear medical violations. In these cases, the model displays high confidence because the context and patterns of the misinformation are too different from the evidence-based tone of "True" health information. However, it is more challenging for the model to identify "Hard" claims that attempt to mimic a professional tone. These claims often use medical terminology and formal syntax to fake a sense of scientific credibility. As shown in Figure 5 below, the model’s performance for complex examples varies. While the model correctly identified one of the claims as "Fake" with 96.92% confidence despite its professional framing, it misclassified the other claim as "True".

Claim Text	Label	Model Prediction	Model Confidence
“Drinking water can cure COVID”	Fake	Fake	99.98%
"Emerging observational data suggests that certain dietary supplements may provide better long-term protection than standard prophylactic interventions..."	Fake	Fake	96.92%
"Research indicates that asymptomatic individuals may not contribute significantly to viral transmission, suggesting that social distancing protocols may be less impactful..."	Fake	True	0.44%

Figure 5: Classification Report for the Final Test Set.

These results illustrate that while the MisinfoLSTM successfully identifies the vast majority of complex claims, it occasionally encounters "blind spots" where the context and structure of the claim are convincing enough to bypass the classification logic. Overall, the model's ability to correctly categorize the majority of these difficult claims demonstrates that it is beyond simple keyword matching and has developed a deeper understanding of medical discourse.

Evaluate Model on New Data

The primary dataset used to train the model consists of information from 5–6 years ago, which was a period heavily dominated by topics related to COVID-19. It is essential to test the model against modern misinformation trends since there are differences in tone, techniques, and subject matter. To ensure the model can generalize to recent data, the final test was conducted using medical claims and tweets that were collected in the past year (8). This new data was entirely unseen by the model during training, validation, or initial testing so it will provide a true measure of its real-world performance. The model maintained strong generalization and achieved an accuracy of 96.30%, which confirms that it has successfully learned underlying linguistic patterns of misinformation rather than relying on time-specific keywords and features.

Discussion

In my initial experimentation, I attempted to train the model on a broader spectrum of general misinformation instead of focusing on medical claims. This approach resulted in poor model performance. I was surprised to learn that while distinct misinformation categories can share similar techniques such as clickbait structures and emotional triggering, their semantic foundations are too different. The model struggled to converge because the vocabulary and contextual nuances for different categories are too distinct. The noise caused the LSTM to not capture meaningful patterns and resulted in terrible accuracy.

The decision to restrict the data exclusively to medical misinformation was the turning point of this project. When focusing on a specific topic, the model could prioritize learning the linguistic patterns and terminology specific to medical claims. With this change, the final model achieved high test accuracy and great quantitative results. I learned that a model's performance is not only a result of its architecture, but also of the thematic consistency and quality of the data it consumes.

Ethical Considerations

Misinformation detection systems introduce ethical challenges such as misclassification harm, algorithmic bias, and data privacy. Falsely labelling accurate medical advice as "fake" generates false positives that can censor valid health discussions. This can lead to backlash from social media users who feel their posts are being unfairly censored (11). Failing to detect harmful misinformation directly deters families from seeking medical treatment. Beyond classification errors, there is a risk of algorithmic bias if the training data overrepresents specific demographics. Finally, because the model processes public social media interactions, it is important to protect user privacy. To mitigate these concerns, the project will utilize diverse training data to ensure balanced and accurate detection of misinformation to successfully protect families from dangerous health decisions without compromising user privacy.

Project Difficulty/Quality

Detecting misinformation is a uniquely challenging problem that is more complex than standard classification tasks. Misinformation rarely follows straightforward patterns or relies on obvious keywords, making it more difficult to analyze. Distinguishing real information from subtle lies requires capturing deep semantic context. This can be challenging with short claims since there are fewer linguistic signals to evaluate.

To tackle this, I developed a custom MisinfoLSTM architecture to map these complex contextual patterns. I researched and integrated optimization techniques such as Dropout for regularization and weight decay to ensure the model effectively weighs nuanced context and does not rely heavily on surface-level features. This pipeline improved the model from a basic classifier into an optimized system that can distinguish misinformation from evidence-based medical claims with high precision.

References

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11)

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